

Bayesian Inference: From Priors to Posteriors

An Intuitive Introduction for Quantitative Researchers

Prepared for Graduate Seminar

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Agenda

- Why Bayesian Inference?
- Bayes' Theorem: The Core Equation
- From Likelihood $\times$ Prior to Posterior
- Conjugate Priors & Analytical Examples
- Beta-Binomial Model in Depth
- Gaussian Inference
- Markov Chain Monte Carlo (MCMC) Overview
- Practical Advantages & Limitations
- Summary & Takeaways

Frequentist vs. Bayesian Thinking

Frequentist Paradigm

- Parameters θ are fixed (unknown constants)
- Data X are random
- Inference via sampling distributions
- Confidence intervals, p -values
- “Probability” = long-run frequency

Bayesian Paradigm

- Parameters θ are random variables
- Data X are fixed (observed)
- Incorporate prior knowledge
- All inference via probability statements
- “Probability” = degree of belief

Bayesian = Updating beliefs with data

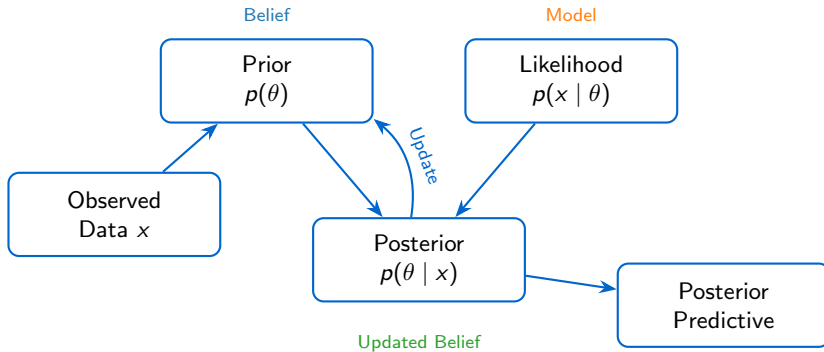
Bayes' Theorem

$$\underbrace{P(\theta | x)}_{\text{Posterior}} = \frac{\underbrace{P(x | \theta)}_{\text{Likelihood}} \times \underbrace{P(\theta)}_{\text{Prior}}}{\underbrace{P(x)}_{\text{Marginal Likelihood}}}$$

- $P(\theta | x)$: Updated belief about θ after seeing data x
- $P(x | \theta)$: Probability of data given parameter (sampling model)
- $P(\theta)$: Prior belief *before* seeing data
- $P(x)$: Normalizing constant $\int P(x | \theta)P(\theta) d\theta$

"Posterior $\propto$ Likelihood $\times$ Prior"

The Bayesian Workflow (TikZ Diagram)



Updating is iterative: today's posterior is tomorrow's prior.

Conjugate Priors

Definition

A prior $p(\theta)$ is *conjugate* to a likelihood $p(x \mid \theta)$ if the posterior $p(\theta \mid x)$ belongs to the same family as the prior.

Example

Likelihood	Conjugate Prior
Binomial / Bernoulli	Beta
Poisson	Gamma
Normal (known σ^2)	Normal
Multinomial	Dirichlet
Exponential	Gamma

Advantage: Closed-form posterior updates – no numerical integration needed.

Beta-Binomial Model (Classic Example)

Coin Flipping with Uncertainty

- Data: y heads in n flips
- Likelihood: $\text{Binomial}(n, \theta)$
- Prior: $\theta \sim \text{Beta}(\alpha, \beta)$
- **Posterior:**
 $\theta \mid y \sim \text{Beta}(\alpha + y, \beta + n - y)$

After 6 heads in 10 flips with prior $\text{Beta}(2, 5)$: posterior = $\text{Beta}(8, 9)$

Density



Posterior Interpretation

With Posterior $p(\theta | x)$ You Can Directly Compute

- Point estimates: posterior mean, median, mode
- Credible intervals: e.g., 95% probability that $\theta \in [a, b]$
- Probability of hypotheses: $P(\theta > 0.5 | x) = \int_{0.5}^1 p(\theta | x) d\theta$
- Predictive distributions for new data

Posterior contains *all* information needed for inference.
Everything follows from probability calculus.

Gaussian Inference with Known Variance

$$\begin{aligned}x_i &\sim \mathcal{N}(\theta, \sigma^2) && \text{(known } \sigma^2\text{)} \\ \theta &\sim \mathcal{N}(\mu_0, \tau_0^2) && \text{(prior)}\end{aligned}$$

Posterior is also Gaussian:

$$\theta \mid \mathbf{x} \sim \mathcal{N}(\mu_n, \tau_n^2)$$

where

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \quad \mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2} \right).$$

Precision (1/variance) of posterior = prior precision + data precision.

When Closed Forms Are Not Available: MCMC

Goal

Draw samples $\{\theta^{(1)}, \theta^{(2)}, \dots, \theta^{(M)}\}$ from the posterior $p(\theta | x)$ so that any integral

$$\mathbb{E}[g(\theta) | x] = \int g(\theta) p(\theta | x) d\theta$$

can be approximated by

$$\frac{1}{M} \sum_{m=1}^M g(\theta^{(m)}).$$

Markov Chain Monte Carlo

- Metropolis-Hastings
- Hamiltonian Monte Carlo (HMC)
- Gibbs Sampling
- No-U-Turn Sampler (NUTS)

Practical Advantages of Bayesian Approach

- Coherent incorporation of prior information
- Direct probabilistic statements (no “ p -values”)
- Natural uncertainty quantification
- Hierarchical modeling is straightforward
- Model comparison via marginal likelihood or LOO-CV
- Works elegantly with small samples

Common Criticisms

- Choice of prior can be subjective (use weakly informative or hierarchical priors)
- Computationally intensive for complex models (but improving rapidly)
- Requires careful model checking (posterior predictive checks)

Bayesian Inference = Learning from data by updating beliefs

- Bayes' theorem is the engine
- $\text{Prior} \times \text{Likelihood} \rightarrow \text{Posterior}$
- Conjugate families give analytical solutions
- MCMC allows arbitrary models
- Posterior contains *everything* you need for decisions under uncertainty

“All statistical inference is updating of beliefs.”
– paraphrasing Lindley & many others

Thank You!

Questions?

All diagrams created with TikZ. Presentation is fully self-contained.